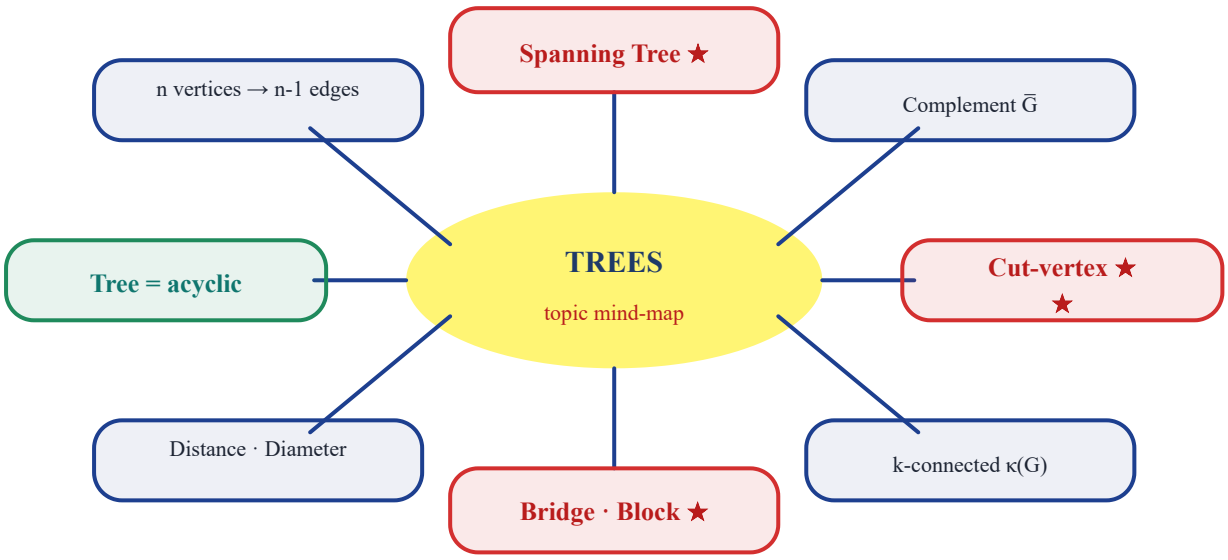
Trees & Graph Connectivity

Tree · Spanning tree · Distance · Complement · Cut-vertex · Bridge · k-connected: exam-ready in 10 sheets



↑ the whole topic on one page: red = high-yield, green = the one definition every question builds from

Syllabus coverage

The syllabus line is "**Graph theory: basic definitions — simple graph, directed graph, subgraph, degree, walk, path, cycle, special graphs (K_n, C_n, $K_{m,n}$), representations (adjacency matrix, adjacency list)**". This lesson covers Trees, Spanning Trees, Distance, Complementary Graphs, Cut-Vertex, Bridges, k-Connected which build on the graph fundamentals.

High-yield list

① Tree: connected + acyclic
② n vertices → n−1 edges
③ Unique path between any two
④ Spanning tree = |V|−1 edges
⑤ dist(u,v) = shortest path
⑥ $G \cup \bar{G} = K_n$
⑦ Cut-vertex: removal ↑ comps
⑧ Bridge: not in any cycle
⑨ $\kappa \le \lambda \le \delta$ (Whitney)
⑩ $\kappa(K_n) = n-1$

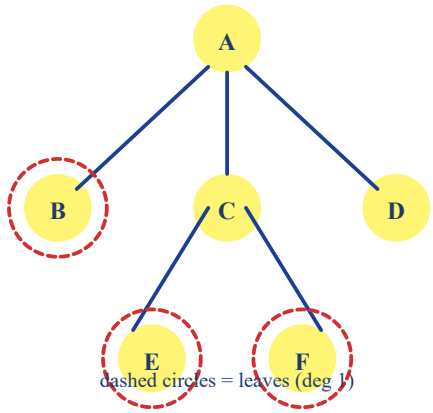
Legend: **yellow** = key term · **red circle** = must-memorise fact · **purple box** = memory trick · **red box** = exam trap · **green** = worked example

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✓ Tree definition & properties … p. 2
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1 Trees

Definition: A **tree** is a **connected, acyclic** undirected graph. Equivalently: a graph with no cycles in which every pair of vertices is joined by exactly one path. A **forest** is a disjoint union of trees — an acyclic graph whose components are trees.



A tree with 6 vertices and 5 edges; B, E, F are leaves

Property	Description
Edges	If a tree has n vertices, it has **n − 1** edges
Unique path	Exactly one path between any two vertices
Adding edge	Adding any edge creates **exactly one cycle**
Removing edge	Removing any edge **disconnects** the graph
Leaves	Every tree with $n \geq 2$ has at least **2 leaves** (deg 1)
Forest	n vertices, k trees → **n − k** edges

Memory trick: "**A tree never loops — it only branches.**" No cycles, no shortcuts, just one road between any two nodes. That is why trees are the skeleton of every network: **minimum edges for maximum reach.**

Exam trap: (i) A tree is **not** just "acyclic" — it must also be **connected**. An acyclic graph with 4 vertices and 2 edges is a forest, not a tree. (ii) A single vertex (n = 1) is a tree with 0 edges — the formula n − 1 still holds. (iii) "Leaves = degree 1" — the root of a rooted tree may have degree 1 and still count as a leaf in the underlying unrooted tree.

Worked example ✍ verify tree properties: A graph has 8 vertices and 7 edges. Is it necessarily a tree? **No** — it could be disconnected (e.g. a triangle + 5 isolated vertices has 3 edges, not 7). A connected graph with n vertices and n − 1 edges **is** a tree. Here: if we confirm connectivity, then yes — 8 vertices, 7 edges, connected ⟹ tree.

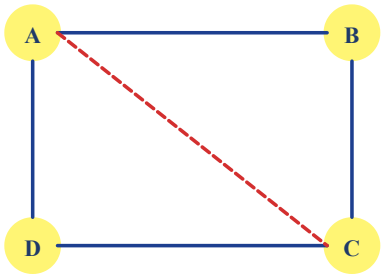
2 Spanning Trees

Definition: A **spanning tree** of a connected graph G = (V, E) is a subgraph T = (V, E') such that: (i) T includes **all vertices** of G, and (ii) T is a **tree** (connected and acyclic). Any spanning tree has exactly **|V| − 1** edges.

Key facts:

- ✓ Every connected graph has **at least one** spanning tree
- ✓ A graph with cycles can have **many** spanning trees
- ✓ Number of edges in any spanning tree = $|V| - 1$
- ✓ Removing one edge from each cycle in G yields a spanning tree

Cayley's formula: The number of distinct **labelled** spanning trees of the complete graph K_n is $\mathbf{n^{n-2}}$. For K_4 this gives $4^2 = \mathbf{16}$ spanning trees. For K_5 this gives $5^3 = \mathbf{125}$.



dashed = one possible removal

C_4: removing any one of 4 edges gives a spanning tree; 4 total

Worked example ✍️ spanning trees of C_4: A cycle C_4 has 4 vertices and 4 edges. Removing any one edge breaks the cycle while keeping the graph connected, giving a spanning tree. Since there are 4 edges, C_4 has exactly **4** spanning trees. For K_4 (6 edges), Cayley gives $4^2 = \mathbf{16}$ labelled spanning trees.

Exam trap: (i) A spanning tree of G is **not** a subgraph of every spanning tree — it is a spanning tree **of G**. Different spanning trees share vertices but may differ in edges. (ii) A disconnected graph has **no** spanning tree — only a **spanning forest**. (iii) K_4 has $4^{4-2} = \mathbf{16}$ spanning trees, not 6. K_5 has $5^3 = 125$.

Memory trick: "**Spanning tree = skeleton.**" It touches every vertex (spans all) and uses the fewest possible edges (tree) to keep everything connected. Think of it as the minimum-cable wiring to link all computers in a network.

IT link: Minimum-spanning-tree algorithms (Kruskal's, Prim's) are used in network design, cluster analysis, and approximating the Traveling Salesman Problem.

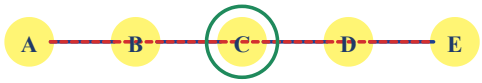
3 Distance in Graphs

Definition: The **distance** d(u, v) between two vertices u and v is the length (number of edges) of a **shortest path** connecting them. If no path exists, $d(u, v) = \infty$.

Property	Description
$d(u, v) = 0$	If and only if $u = v$
$d(u, v) = d(v, u)$	Symmetric in undirected graphs
Triangle inequality	$d(u, w) \leq d(u, v) + d(v, w)$
Infinite distance	If u and v are in different components

Derived metrics:

- ✓ **Eccentricity:** $ecc(v) = \max\{d(v, u) \mid u \in V\}$
- ✓ **Diameter: diam(G)** $= \max\{ecc(v) \mid v \in V\}$
- ✓ **Radius: rad(G)** $= \min\{ecc(v) \mid v \in V\}$
- ✓ **Center:** $\{v \mid ecc(v) = rad(G)\}$
- ✓ **Periphery:** $\{v \mid ecc(v) = diam(G)\}$



green circle = center vertex C

P_5: ecc(A)=4, ecc(B)=3, ecc(C)=2, ecc(D)=3, ecc(E)=4; diam=4, rad=2

Worked example ✍ P_5 path graph: Vertices A–B–C–D–E in a line. Distances from A: d(A,B)=1, d(A,C)=2, d(A,D)=3, d(A,E)=4, so ecc(A)=4. Similarly ecc(B)=3, ecc(C)=2, ecc(D)=3, ecc(E)=4. **Diameter** = max ecc = **4**. **Radius** = min ecc = **2**. **Center** = {C}. **Periphery** = {A, E}.

Exam trap: (i) Diameter is the **maximum** eccentricity, not the maximum degree. (ii) The center may contain **one or two adjacent** vertices — e.g. P_4 has center {B, C}. (iii) In a disconnected graph, distance is ∞ between components, so diameter is often defined only for connected graphs.

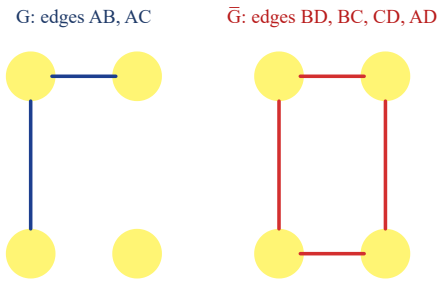
Worked example ✍ star graph S_4: Center vertex c joined to leaves a, b, d. Distances: d(c,a)=d(c,b)=d(c,d)=1, d(a,b)=d(a,d)=d(b,d)=2. Eccentricities: ecc(c)=1, ecc(a)=ecc(b)=ecc(d)=2. **Diameter** = 2, **Radius** = 1, **Center** = {c}, **Periphery** = {a, b, d}. Note: radius ≠ diameter here — only paths and cycles can have rad = diam/2 approximately.

4 Complementary Graphs

Definition: The **complement** of a graph G = (V, E), denoted $\bar{G}$, has the **same vertex set** V, and an edge {u, v} is in $\bar{G}$ **if and only if** {u, v} is **not** in E. Thus G and $\bar{G}$ partition the edges of the complete graph K_n.

Property	Description
$G \cup \bar{G}$	Complete graph $\mathbf{K_n}$ (edge-disjoint union)
$\bar{\bar{G}}$	G (complement of complement is original)
Edges	$\lvert E(\bar{G})\rvert = \mathbf{n(n-1)/2 - \lvert E(G)\rvert}$
Self-compl.	$G \cong \bar{G}$; requires $n \equiv 0$ or $1 \pmod 4$

Self-complementary: A graph isomorphic to its complement. Example: P_4 (path on 4 vertices) is self-complementary. For a self-complementary graph on n vertices, $|E| = \mathbf{n(n-1)/4}$. A self-complementary graph on 8 vertices has $8 \cdot 7/4 = \mathbf{14}$ edges.



$G \cup \bar{G} = K_4$ (6 edges total)

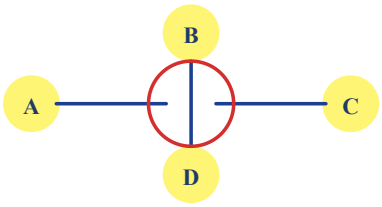
G has 2 edges, $\bar{G}$ has 4; together they form K_4 with 6 edges = $4 \cdot 3/2$

Exam trap: (i) G and $\overline{G}$ **never** share an edge — their edge sets are complementary within K_n. (ii) Both G and $\overline{G}$ cannot be disconnected for $n \geq 2$ — at least one is connected. (iii) A self-complementary graph on n vertices has exactly $n(n-1)/4$ edges, so $n(n-1)$ must be divisible by 4 — this means $n \equiv 0$ or $1 \pmod 4$.

Memory trick: "**Complement = the rest.**" Take K_n, remove G's edges, what remains is $\overline{G}$. For a self-complementary graph, G and $\overline{G}$ are mirror images in edge space — each gets exactly half of K_n's edges.

5 Cut-Vertices (Articulation Points)

Definition: A vertex v is a **cut-vertex** (articulation point) if removing v and all incident edges **increases** the number of connected components. A graph with no cut-vertices is **2-connected** (or non-separable).



B is a cut-vertex (red circle)

Removing B leaves {A}, {C}, {D} — 3 components; B is a cut-vertex

Key properties:

- ✓ Bridges and cut-vertices are **weak points** in networks
- ✓ Leaves are **not** cut-vertices (except in K_2)
- ✓ Every cycle has **no** cut-vertices
- ✓ A block is a maximal subgraph **without** cut-vertices
- ✓ A tree with $n \geq 3$ has $n - 2$ cut-vertices (all non-leaf vertices)

Exam trap: A leaf (degree-1 vertex) is **never** a cut-vertex in a tree with ≥ 3 vertices — removing it leaves the rest still connected. Only in K_2 does the single edge's endpoint count as a cut-vertex. Also: a graph with a cut-vertex has $\kappa(G) = 1$, so it is **not** 2-connected.

Memory trick: "**A cut-vertex is a bottleneck.**" Remove it and the network falls apart into more pieces. Think of a bridge city on a river — close it and the two banks are isolated.

Worked example ✍ find cut-vertices: Graph: A—B—C with D attached to B (B is centre). Removing B gives components {A}, {C}, {D} — **3 components** (was 1). So B is a cut-vertex. A, C, D are leaves — removing any one leaves 2 vertices still connected, so they are **not** cut-vertices. In a path P_5 (A–B–C–D–E), the cut-vertices are B, C, D (3 of them); A and E are leaves, not cut-vertices.

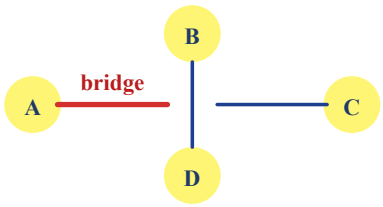
IT link: Cut-vertices identify single points of failure in communication networks, power grids, and road systems — locations where one outage isolates entire regions.

6 Bridges (Cut-Edges)

Definition: An edge e is a **bridge** (cut-edge) if removing e **increases** the number of connected components. Equivalently: e is a bridge **if and only if** it belongs to **no cycle** in G.

Key facts:

- ✓ Every edge in a **tree** is a bridge
- ✓ An edge is a bridge ⟺ it is not in any cycle
- ✓ Removing a bridge **disconnects** the graph
- ✓ Bridges are the edge-analogue of cut-vertices
- ✓ A graph with no bridges is **2-edge-connected**



every edge of a tree is a bridge

A—B is a bridge — like every edge of this tree

DFS bridge-finding algorithm: Compute **discovery time** disc[u] and **low-link** value low[u] = min(discover time of u, discover time of ancestor reachable via back-edge from subtree of u). Edge (u, v) (u = parent of v) is a **bridge** if **low[v] > disc[u]**.

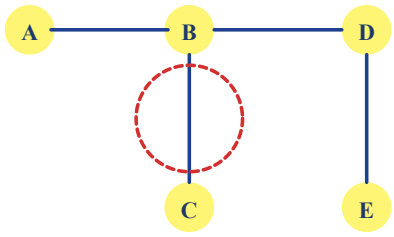
Exam trap: (i) An edge in a **cycle** is **never** a bridge — removing one edge from a cycle leaves a path, still connected. (ii) **Every tree edge is a bridge**, but not every bridge is in a tree — bridges exist in graphs with cycles too. (iii) low[v] > disc[u] (strictly greater); low[v] = disc[u] means there is a back-edge, so (u,v) is **not** a bridge.

Memory trick: "**No cycle = bridge.**" If you can 绕 (go around) from v back to u without using edge (u,v), then (u,v) is safe — it is not a bridge. If not, it is a bridge.

Worked example ✍️ bridges in a triangle with a tail: Graph: A—B—C forms a triangle (A-B, B-C, C-A), and C—D is a tail. Edge C—D is a bridge (removing it isolates D). Edges A-B, B-C, C-A are **not** bridges — each lies on the triangle cycle. Removing any one triangle edge leaves a path A-B-C, still connected. So this graph has exactly **1 bridge**: C—D.

7 Blocks & k-Connected Graphs

Block: A **block** is a maximal connected subgraph with **no cut-vertex**. Blocks are either: (i) a maximal 2-connected subgraph, (ii) a bridge (single edge that is a block), or (iii) an isolated vertex. A **block tree** (BC-tree) has nodes for blocks and cut-vertices, with edges linking each cut-vertex to the blocks it belongs to.



B is a cut-vertex; blocks: {A,B}, {B,C,D}, {D,E}

Block decomposition: three blocks joined at cut-vertices B and D

k-connected: A graph is **k-connected** if $|V| > k$ and removing any set of fewer than k vertices leaves the graph connected. The **vertex connectivity** $\kappa(G)$ is the minimum number of vertices whose removal disconnects G (or makes it trivial).

Edge connectivity: The **edge connectivity** $\lambda(G)$ is the minimum number of edges whose removal disconnects G. **Whitney's Theorem: $\kappa(G) \leq \lambda(G) \leq \delta(G)$**, where $\delta(G)$ is the minimum degree.

Exam trap: (i) $\kappa(K_n) = n - 1$, **not** n. You need to remove all but one vertex to disconnect K_n. (ii) A graph with a cut-vertex has $\kappa = 1$ — it is **not** 2-connected. (iii) $\kappa \leq \lambda \leq \delta$ is always true; equality on both sides holds for K_n and C_n but not generally.

Memory trick: **"κ is the weakest link, δ is the strongest ceiling."** Vertex connectivity (κ) ≤ edge connectivity (λ) ≤ minimum degree (δ). The minimum degree is an upper bound — you can always disconnect by removing all neighbours of a min-degree vertex, but sometimes fewer suffice.

Worked example ✍ connectivity values: (a) Tree with $n \geq 2$: $\kappa = 1$ (has cut-vertices), $\lambda = 1$ (has bridges). (b) Cycle C_n, $n \geq 3$: $\kappa = 2$, $\lambda = 2$ — removing any 1 vertex or edge leaves a path (connected). (c) K_6: $\kappa = 5$, $\lambda = 5$ — you must remove 5 vertices/edges to isolate one vertex. (d) $K_{3,3}$: $\kappa = 3$, $\lambda = 3$ (3-regular, each vertex has degree 3).

Night-Before-the-Exam Sheet

Fact sheet:

Tree: connected + acyclic	n vertices → n − 1 edges
Unique path between any 2 vertices	Adding edge → exactly one cycle
Removing edge → disconnects	Leaves: at least 2 when $n \geq 2$
Forest: n vertices, k trees → n − k edges	Spanning tree: $\|V\| - 1$ edges
Cayley: K_n has n^{n-2} spanning trees	Distance: shortest-path length
Diameter = max eccentricity	Radius = min eccentricity
Center: vertices with ecc = rad	Periphery: vertices with ecc = diam
$G \cup \bar{G} = K_n$ (edge-disjoint)	$\|E(\bar{G})\| = n(n-1)/2 - \|E(G)\|$
Self-complement: $n \equiv 0,1 \pmod 4$	Cut-vertex: removal ↑ components
Bridge: edge not in any cycle	Bridge test: low[v] > disc[u]
Block: maximal subgraph, no cut-v	$\kappa(G) \leq \lambda(G) \leq \delta(G)$ (Whitney)
$\kappa(K_n) = n - 1$; $\kappa(C_n) = 2$	k-connected: survives removal of k−1 verts

Top 12 one-liners for MCQs

1. Tree: n vertices, n − 1 edges
2. Forest: n vertices, k trees → n − k edges
3. Spanning tree of G has $|V| - 1$ edges
4. Cayley: K_n has n^{n-2} spanning trees
5. d(u,v) = length of shortest path
6. diam(G) = max ecc; rad(G) = min ecc
7. $|E(\bar{G})| = n(n-1)/2 - |E(G)|$
8. Cut-vertex: removal increases components
9. Bridge: edge not in any cycle
10. Bridge test: low[v] > disc[u]
11. Block: maximal subgraph with no cut-v
12. Whitney: $\kappa \leq \lambda \leq \delta$

Top traps: "Tree" requires both connected AND acyclic — acyclic alone is a forest · A leaf is never a cut-vertex in $n \geq 3$ · low[v] > disc[u] is strict — equality means a back-edge exists · Self-complementary needs $n(n-1)/4$ edges, so $n \equiv 0$ or $1 \pmod 4$ · $\kappa(K_6) = 5$, not 6 · Diameter is max eccentricity, not max degree · Center of P_4 is {B,C}, not a single vertex

All tricks together: "A tree never loops — it only branches" · "Spanning tree = skeleton" · "Complement = the rest" · "A cut-vertex is a bottleneck" · "No cycle = bridge" · "κ is the weakest link, δ is the strongest ceiling".

✓ Practice MCQs (exam pattern, 10 marks)

1. How many edges does a tree with 20 vertices have?

(a) 20 (b) 21

(c) 18 (d) 19

2. Which is a bridge of a graph?

(a) An edge joining a vertex to itself
(b) An edge between max-degree vertices
(c) An edge on every cycle
(d) An edge whose removal ↑ components

3. G has 8 vertices and 10 edges. How many edges does $\overline{G}$ have?

(a) 10
(b) 28
(c) 38
(d) 18

4. A forest has 15 vertices and 4 trees. How many edges?

(a) 14
(b) 11
(c) 15
(d) 19

5. How many distinct labeled spanning trees does the complete graph K_4 have?

(a) 4
(b) 12
(c) 16
(d) 64

6. Which is true for every tree with ≥ 2 vertices?

(a) It contains a cycle
(b) Every vertex is a cut-vertex
(c) It has exactly n edges
(d) Every edge is a bridge

7. What is $\kappa(K_6)$, the vertex connectivity of K_6?

(a) 6
(b) 5
(c) 4
(d) 15

8. For path graph P_7, what are the radius and diameter?

(a) 3 and 6
(b) 6 and 3
(c) 3.5 and 7
(d) 4 and 6

9. For any graph G, which inequality (Whitney's) always holds?

(a) $\lambda \le \kappa \le \delta$
(b) $\delta \le \kappa \le \lambda$
(c) $\kappa \le \lambda \le \delta$
(d) $\kappa \le \delta \le \lambda$

10. A self-complementary graph with 8 vertices has how many edges?

(a) 14
(b) 28
(c) 7
(d) 16

Answer key → 1-d · 2-d · 3-d · 4-b · 5-c · 6-d · 7-b · 8-a · 9-c · 10-a

Written question: Prove that every tree with n vertices has exactly n − 1 edges. 10

1. **Base case (n = 1).** A tree with one vertex has 0 edges. Formula: 1 − 1 = 0. ✓ 2
2. **Inductive hypothesis.** Assume every tree with k vertices has k − 1 edges.
3. **Inductive step.** Let T be a tree with k + 1 vertices. T has at least two leaves (vertices of degree 1). Remove a leaf v and its incident edge e. The remaining graph T' has k vertices and is still a tree (removing a leaf preserves connectivity and acyclicity). By the hypothesis, T' has k − 1 edges. Therefore T has (k − 1) + 1 = **k** = (k + 1) − 1 edges. ✓ 5
4. **Conclusion.** By mathematical induction, every tree with n vertices has exactly n − 1 edges for all n ≥ 1. 2